SHANZAY HUSSAIN

Software Engineer | Full-Stack and AI Developer

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Portfolio • LinkedIn • GitHub

EDUCATION

B.E. Software Engineering 2024 – 2028

NED University of Engineering & Technology, Karachi

TECHNICAL SKILLS

Frontend: React.js, React Native, Vite, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, Django, REST APIs

Databases: MySQL, PostgreSQL, MongoDB, ChromaDB

Languages: Python, TypeScript, JavaScript, C++, C#

AI: LLM Integration, Prompt Engineering, RAG, Embeddings, Scikit-learn

Tools & Platforms: Git, GitHub, Vercel, Railway, Linux, n8n, Docker

EXPERIENCE

Backend Intern — *ITVE Pvt Ltd* June 2026 – Aug 2026

- Contributed to the backend of a FastAPI-based REST API with JWT authentication, MongoDB (via Motor) for data persistence, and Cloudinary for image upload handling.

Full Stack Intern — *Solace* Aug 2026 – Sept 2026

- Designed and built a role-based HR dashboard system replacing manual Excel-based reporting, with a React frontend and FastAPI/PostgreSQL backend enforcing access control across admin, city-level head, and employee roles.

PROJECTS

RAG Intelligence Chatbot — **Multimodal AI Knowledge Assistant**

React.js · Vite · TypeScript · FastAPI · MongoDB Atlas · ChromaDB · Firebase Auth · OpenAI (GPT-5.4-mini) · fastembed

- Built a full-stack multimodal RAG chatbot supporting document/image uploads, conversational Q&A, and persistent chat history.
- Developed a ChromaDB-based retrieval pipeline with local embeddings and similarity-threshold routing for document-grounded responses.
- Implemented image understanding and cross-conversation knowledge search, enabling retrieval across previously uploaded documents and images.

SmartBudget — **AI-Powered Personal Finance & Spending Risk Platform**

React.js · Vite · TypeScript · Express.js · Tailwind CSS · Python · FastAPI · scikit-learn · Firebase · Groq AI (LLaMA 3.3 70B) · Chart.js

- Built an end-to-end personal finance platform with budget tracking, analytics, and ML-based overspending risk prediction.
- Developed real-time spending analytics and behavioral insights based on spending trends, mood, and day of the week.
- Integrated an AI financial assistant to generate personalized insights from user spending data.

MediChain — **AI & Blockchain Pharmaceutical Supply-Chain Marketplace**

React Native · FASTAPI · Postgresql · Supabase · QR code · Docker

- Built a mobile-based pharmaceutical marketplace and supply-chain verification platform connecting manufacturers, distributors, pharmacies, and customers.
- Implemented QR-based medicine tracking with a hash-chained ledger to record custody history, detect tampering, and identify suspicious QR-code reuse across the supply chain.
- Developed AI-driven demand forecasting, inventory analysis, and anomaly detection using simulated pharmaceutical sales and supply-chain data to identify potential shortages and unusual transactions.

InterviewIQ — **AI-Powered Mock Interview Platform**

React.js · Vite · Express.js · Supabase · Groq AI (LLaMA 3.3 70B) · Chart.js

- Built a full-stack AI mock interview platform with automated interview interactions and performance tracking.
- Developed an analytics dashboard using Chart.js to visualize performance trends and identify skill gaps.
- Implemented speech recognition to create a more realistic interview experience.